

```
!pip install transformers torch accelerate
```

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⇒ Requirement already satisfied: transformers in /usr/local/lib/python3.10/dist-
Requirement already satisfied: torch in /usr/local/lib/python3.10/dist-package
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Requirement already satisfied: mpmath>=0.19 in /usr/local/lib/python3.10/dist-

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pip install --upgrade gradio
```

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⇒ Requirement already satisfied: gradio in /usr/local/lib/python3.10/dist-pack
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➡ 'en_US.UTF-8'

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➡ /usr/local/lib/python3.10/dist-packages/transformers/models/auto/tokenization_
  warnings.warn(

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(...)okenizer_config.json: 100% 57.4kB/s]

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100% 3.51MB/s]

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```
from transformers import pipeline
import torch
```

```
llama_pipeline = pipeline(
    "text-generation",
    model=model,
    torch_dtype=torch.float16,
    device_map="auto",
)
```

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	(...)of-000002.safetensors: 100%	247MB/s]
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Double-click (or enter) to edit

```

SYSTEM_PROMPT = """<s>[INST] <<SYS>>
You are a helpful bot. Your answers are clear and concise.
<</SYS>>

"""

# Formatting function for message and history
def format_message(message: str, history: list, memory_limit: int = 3) -> str:
    """
    Formats the message and history for the Llama model.

    Parameters:
        message (str): Current message to send.
        history (list): Past conversation history.
        memory_limit (int): Limit on how many past interactions to consider.

    Returns:
        str: Formatted message string
    """
    # always keep len(history) <= memory_limit
    if len(history) > memory_limit:
        history = history[-memory_limit:]

    if len(history) == 0:
        return SYSTEM_PROMPT + f"{message} [/INST]"

    formatted_message = SYSTEM_PROMPT + f"{history[0][0]} [/INST] {history[0][1]}

    # Handle conversation history
    for user_msg, model_answer in history[1:]:
        formatted_message += f"<s>[INST] {user_msg} [/INST] {model_answer} </s>"

    # Handle the current message
    formatted_message += f"<s>[INST] {message} [/INST]"

    return formatted_message

```

```
# Generate a response from the Llama model
def get_llama_response(message: str, history: list) -> str:
    """
    Generates a conversational response from the Llama model.

    Parameters:
        message (str): User's input message.
        history (list): Past conversation history.

    Returns:
        str: Generated response from the Llama model.
    """
    query = format_message(message, history)
    response = ""

    sequences = llama_pipeline(
        query,
        do_sample=True,
        top_k=10,
        num_return_sequences=1,
        eos_token_id=tokenizer.eos_token_id,
        max_length=1024,
    )

    generated_text = sequences[0]['generated_text']
    response = generated_text[len(query):] # Remove the prompt from the output

    print("Chatbot:", response.strip())
    return response.strip()
```

```
import gradio as gr

gr.ChatInterface(get_llama_response).launch()
```

 Setting queue=True in a Colab notebook requires sharing enabled. Setting `share`

Colab notebook detected. To show errors in colab notebook, set debug=True in launch().
Running on public URL: <https://d734c667923c2816bc.gradio.live>

This share link expires in 72 hours. For free permanent hosting and GPU upgrade, see [Gradio Cloud](#).



No interface is running right now

